

12.4 Tree-width

As indicated by Figure 12.3.1, the bags of $(T, \mathcal{V})$ reflect the structure of the tree T , so in this sense the graph G decomposed resembles a tree. However, this is valuable only inasmuch as the structure of G within each part is negligible: the smaller the bags, the closer the resemblance.

This observation motivates the following definition. The *width* of $(T, \mathcal{V})$ is the number

width

$$\max \{ |V_t| - 1 : t \in T \},$$

and the *tree-width* $\text{tw}(G)$ of G is the least width of any tree-decomposition of G . As one easily checks,⁴ trees themselves have tree-width 1.

tree-width
 $\text{tw}(G)$

By Lemmas 12.3.2 and 12.3.3, the tree-width of a graph will never be increased by deletion or contraction:

(12.3.2)
(12.3.3)

Lemma 12.4.1. *If $H \preceq G$ then $\text{tw}(H) \leq \text{tw}(G)$.*

□

[12.6.2]

Graphs of bounded tree-width are sufficiently similar to trees that it becomes possible to adapt the proof of Kruskal's theorem to the class of these graphs; very roughly, one has to iterate the 'minimal bad sequence' argument from the proof of Lemma 12.1.3 $\text{tw}(G)$ times. This takes us a step further towards a proof of the graph minor theorem:

Theorem 12.4.2. (Robertson & Seymour 1990)

[12.7.1]
[12.7.3]

For every integer $k > 0$, the graphs of tree-width $< k$ are well-quasi-ordered by the minor relation.

In order to make use of Theorem 12.4.2 for a proof of the full graph minor theorem, we should be able to say something about the graphs which it does not cover, i.e., to deduce some information about a graph from the assumption that its tree-width is large. Our next result, the *tree-width duality theorem*, achieves just that: it identifies a canonical obstruction to small tree-width, a structural phenomenon that occurs in a graph if and only if its tree-width is large.

Let us say that two subsets of $V(G)$ *touch* if they have a vertex in common or G contains an edge between them. A set of mutually touching connected vertex sets in G is a *bramble*. Extending our terminology of Chapter 2, we say that a subset of $V(G)$ *covers* (or is a *cover* of) a bramble $\mathcal{B}$ if it meets every element of $\mathcal{B}$. The least number of vertices covering a bramble is the *order* of that bramble. A *k-bramble* is one of order k .

touch

bramble

cover

order

⁴ Indeed the '−1' in the definition of width serves no other purpose than to make this statement true.

grid A typical example of a bramble is the set of crosses in a grid. The $k \times k$ *grid* is the graph on $\{1, \dots, k\}^2$ with the edge set

$$\{(i, j)(i', j') : |i - i'| + |j - j'| = 1\}.$$

The *crosses* of this grid are the k^2 sets

$$C_{ij} := \{(i, \ell) \mid \ell = 1, \dots, k\} \cup \{(\ell, j) \mid \ell = 1, \dots, k\}.$$

Thus, the cross C_{ij} is the union of the grid's i th row and its j th column. Clearly, the crosses of the $k \times k$ grid form a k -bramble: they are covered by any row or column, while any set of fewer than k vertices misses both a row and a column, and hence a cross.

Theorem 12.4.3. (Seymour & Thomas 1993)

Let $k \geq 0$ be an integer. A graph has tree-width $< k$ if and only if it contains no bramble of order $> k$.

(3.3.1) *Proof.* Let $G = (V, E)$ be a graph. For the forward implication, let $\mathcal{B}$ be
(12.3.1) any bramble in G . We show that every tree-decomposition $(T, (V_t)_{t \in T})$
(12.3.2) of G has a bag that covers $\mathcal{B}$.
(12.3.3)

As in the proof of Lemma 12.3.4 we start by orienting the edges $t_1 t_2$ of T . If $X := V_{t_1} \cap V_{t_2}$ covers $\mathcal{B}$, we are done. If not, then for each $B \in \mathcal{B}$ disjoint from X there is an $i \in \{1, 2\}$ such that $B \subseteq U_i \setminus X$ (defined as in Lemma 12.3.1); recall that B is connected. This i is the same for all such B , because they touch. We now orient the edge $t_1 t_2$ towards t_i .

If every edge of T is oriented in this way, it has a node t all whose incident edges are oriented towards it. Then V_t covers $\mathcal{B}$ – just as in the proof of Lemma 12.3.4.

good For our proof of the converse implication we need the notion of a *good* tree-decomposition $(T, \mathcal{V})$ of G , with $\mathcal{V} = (V_t)_{t \in T}$ say: one where
 $|V_t| \leq k$ for at least one t and $|V_t| \leq k$ whenever t is not a leaf of T .
Then $|V_t| \leq k$ for the unique neighbour t of any leaf x with $|V_x| > k$.
petal The set $V_x \setminus V_t \neq \emptyset$ is the *petal* of such a leaf x with $|V_x| > k$.

Suppose now that $\text{tw}(G) \geq k$, and let us find a bramble of order $> k$. Note that G has a good tree-decomposition – for example, with $T = K^2$ and parts $\emptyset, V$. And since $\text{tw}(G) \geq k$, every good tree-decomposition of G has a petal. Let $\mathcal{B}$ be a minimal set of petals of good tree-decompositions satisfying

- (i) $\mathcal{B}$ contains a petal of every good tree-decomposition;
- (ii) $\mathcal{B}$ is closed under taking supersets among petals: if $X \subseteq X'$ are both petals of good tree-decompositions and $X \in \mathcal{B}$, then $X' \in \mathcal{B}$.

Let $\mathcal{B}' := \{X \in \mathcal{B} \mid G[X] \text{ is connected}\}$. To complete our proof we show that $\mathcal{B}'$ is a bramble of order $> k$.

Let us show first that no set Z of at most k vertices covers $\mathcal{B}'$. As $\text{tw}(G) \geq k$ we have $V \setminus Z \neq \emptyset$, and if $C_1, \dots, C_n$ are the components of $G - Z$, then G has a tree-decomposition into Z and the sets $V(C_i) \cup Z$, whose decomposition tree is a star with Z as its central bag. This is a good tree-decomposition, so by (i) it has a petal in $\mathcal{B}$. But any such petal is one of the C_i , so it is connected and thus lies in $\mathcal{B}'$. Hence Z fails to cover $\mathcal{B}'$, as claimed.

It remains to show that every two sets in $\mathcal{B}'$ touch. In fact, let us show that every two sets in $\mathcal{B}$ touch. If not, we can find $X, Y \in \mathcal{B}$ that do not touch and are $\subseteq$ -minimal in $\mathcal{B}$. Since $\mathcal{B} \setminus \{X\}$ and $\mathcal{B} \setminus \{Y\}$ still satisfy (ii), the minimality of $\mathcal{B}$ implies that they violate (i). So there exists a good tree-decomposition $(T_1, \mathcal{V}_1)$ whose only petal in $\mathcal{B}$ is X , and a good tree-decomposition $(T_2, \mathcal{V}_2)$ whose only petal in $\mathcal{B}$ is Y . Let us choose T_1 and T_2 disjoint.

Let X be the petal of $x \in T_1$, and Y the petal of $y \in T_2$. By Lemma 12.3.1, x is unique in T_1 and y is unique in T_2 . Deleting any vertices outside $N(X)$ and $N(Y)$ from the adhesion sets of x and y , respectively, we may assume that their bags are exactly $X \cup N(X)$ and $Y \cup N(Y)$. (The trimmed bags still have size $> k$, since otherwise the modified tree-decomposition would have no petal in $\mathcal{B}$.) Since any petal in $(T_1, \mathcal{V}_1)$ containing Y would lie in $\mathcal{B} \setminus \{X\}$, by (ii), there is no such petal. Similarly, no petal in $(T_2, \mathcal{V}_2)$ contains X .

To complete our proof we show the following:

There is a good tree-decomposition $(T, \mathcal{V})$ all whose petals are contained in petals of $(T_1, \mathcal{V}_1)$ and $(T_2, \mathcal{V}_2)$ and in which neither X nor Y is a petal. (*)

Since X and Y are the only petals of $(T_1, \mathcal{V}_1)$ and $(T_2, \mathcal{V}_2)$ in $\mathcal{B}$ and have no proper subsets in $\mathcal{B}$, the $(T, \mathcal{V})$ from (*) will have no petal in $\mathcal{B}$, by (ii) for $\mathcal{B}$. As this contradicts (i), our proof will then be complete.

So let us prove (*). As X and Y do not touch, the set $N(X)$ is disjoint from both X and Y and separates them in G . Hence G has a separation $\{A, B\}$ such that $X \subseteq A \setminus B$ and $Y \subseteq B \setminus A$. As $|N(X)| \leq k$ since X is a petal, choosing $\{A, B\}$ of minimum order ensures that $S := A \cap B$ has size at most k . By the minimality of S and Menger's Theorem 3.3.1, there is a family $\{P_s \mid s \in S\}$ of disjoint S - $N(X)$ paths in $G[A]$ and a family $\{Q_s \mid s \in S\}$ of disjoint S - $N(Y)$ paths in $G[B]$.

Let H be the minor of G obtained by deleting $A \setminus \bigcup_{s \in S} V(P_s)$ and contracting each of the paths P_s . Identifying the contracted branch sets $V(P_s)$ with their representatives s , we may think of H as obtained from $G[B]$ by adding some edges on S . Let $(T_1, \mathcal{V}_1')$ be the tree-decomposition which $(T_1, \mathcal{V}_1)$ induces on H as in Lemmas 12.3.2 and 12.3.3, and think of it as a tree-decomposition of $G[B]$. Thus for any $t \in T_1$, with a

bag $V_t^1 \in \mathcal{V}_1$, say, its bag V_t in $\mathcal{V}'_1$ is

$$V_t = (V_t^1 \cap B) \cup \{s \in S \mid V_t^1 \cap V(P_s) \neq \emptyset\} \quad (1)$$

(Fig. 12.4.1). In particular, $V_x = S$, since $V_x = X \cup N(X) \subseteq A$ and $N(X)$ meets every P_s . Similarly, let J be the minor of G obtained by deleting $B \setminus \bigcup_{s \in S} V(Q_s)$ and contracting the paths Q_s , and let $(T_2, \mathcal{V}'_2)$ be the tree-decomposition which $(T_2, \mathcal{V}_2)$ induces on J . As before, think of this as a tree-decomposition of $G[A]$ in which S is the bag corresponding to y .

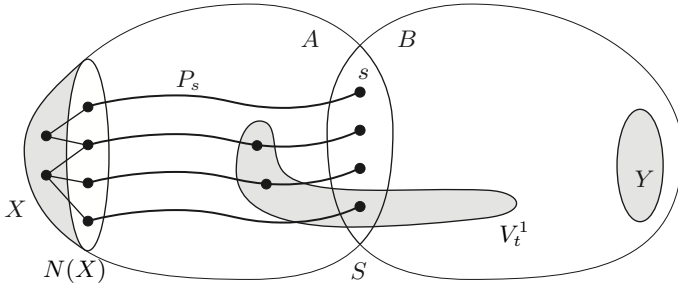


Fig. 12.4.1. To obtain V_t from $V_t^1 \cap B$, add two vertices from S

Let T be obtained from the (disjoint) trees T_1 and T_2 by identifying x and y into a new node r . As Y and X are non-empty, x is not the only node of T_1 and y is not the only node of T_2 , so r is not a leaf of T . Let $V_r := S$. For all $t \in T - r$ let V_t be the bag in $\mathcal{V}'_1$ or $\mathcal{V}'_2$ that corresponds to t there, thought of as a subset of B if $t \in T_1$, or of A if $t \in T_2$. We claim that $(T, \mathcal{V})$ with $\mathcal{V} = (V_t)_{t \in T}$ is a good tree-decomposition of G satisfying $(*)$.

Using that $(T_1, \mathcal{V}'_1)$ and $(T_2, \mathcal{V}'_2)$ are tree-decompositions of $G[B]$ and $G[A]$, it is easy to check that $(T, \mathcal{V})$ is indeed a tree-decomposition of G . The non-leaves of T are precisely those of T_1 and T_2 , plus r . We have already seen that $|S| \leq k$. For $t \in T_1 - x$, its bag V_t in $\mathcal{V}$ is no larger than its bag V_t^1 in $\mathcal{V}_1$: by (1), there exists for every $s \in V_t \setminus V_t^1$ a vertex of P_s in $V_t^1 \setminus V_t$. Similarly, the bags V_t with $t \in T_2$ are no larger than their corresponding bags in $\mathcal{V}_2$. Thus, $(T, \mathcal{V})$ is good.

To show that $(T, \mathcal{V})$ satisfies $(*)$, consider a petal Z in $(T, \mathcal{V})$ of a leaf z of T . Then z is a leaf also of T_1 or T_2 . Let us assume that $z \in T_1$; then $V_z \subseteq B$. By axiom (T3) for $(T, \mathcal{V})$, any vertex of $V_z \cap V_r$ lies in the adhesion set of V_z , so $Z \cap S = \emptyset$. Hence $Z \subseteq B \setminus A$. But this implies that Z lies inside the petal Z_1 of z in $(T_1, \mathcal{V}_1)$: recall that the bag V_z^1 corresponding to z in $\mathcal{V}_1$ has size at least $|V_z| > k$ (so it can have a petal) and differs from V_z only by vertices in A , and likewise for the neighbour of z in T_1 and T . Finally, $Z \neq X$ because $X \subseteq A \setminus B$, and $Z \neq Y$ since the petal $Z_1 \supseteq Z$ of z in $(T_1, \mathcal{V}_1)$ does not contain Y by assumption. This completes the proof of $(*)$, and hence of the theorem. $\square$

Often, Theorem 12.4.3 is stated in terms of the *bramble number* of a graph, the largest order of any bramble in it. The theorem then says that the tree-width of a graph is exactly one less than its bramble number.

How useful even the easy forward implication of Theorem 12.4.3 can be is exemplified once more by our example of the crosses bramble in the $k \times k$ grid: this bramble has order k , so by the theorem the $k \times k$ grid has tree-width at least $k - 1$. (Try to show this without the theorem!)

In fact, the $k \times k$ grid has tree-width k (Exercise 34). But more important than its precise value is the fact that the tree-width of grids tends to infinity with their size. For as we shall see, large grid minors pose another canonical obstruction to small tree-width: not only do large grids (and hence all graphs containing large grids as minors; cf. Lemma 12.4.1) have large tree-width, but conversely every graph of large tree-width has a large grid minor (Theorem 12.6.3).

In Section 12.5 we shall place these within the wider framework of *tangles*, another central notion in graph minor theory. Using tangles one can formulate a more general duality theory between highly connected substructures and trees, of which Theorem 12.4.3 is but a special case.

Tree-width can also be expressed as follows:

Proposition 12.4.4. $\text{tw}(G) = \min \{ \omega(H) - 1 \mid G \subseteq H; H \text{ chordal} \}.$

(12.3.2)

Proof. By Corollary 12.3.5 and Proposition 12.3.6, each of the graphs H considered for the minimum has a tree-decomposition of width $\omega(H) - 1$. Every such tree-decomposition induced one of G by Lemma 12.3.2, so $\text{tw}(G) \leq \omega(H) - 1$ for every H .

(12.3.5)

(12.3.6)

Conversely, let us construct an H as above with $\omega(H) - 1 \leq \text{tw}(G)$. Let $(T, \mathcal{V})$ be a tree-decomposition of G of width $\text{tw}(G)$. For every $t \in T$ let K_t denote the complete graph on V_t , and put $H := \bigcup_{t \in T} K_t$. Clearly, $(T, \mathcal{V})$ is also a tree-decomposition of H . By Proposition 12.3.6, H is chordal, and by Corollary 12.3.5, $\omega(H) - 1$ is at most the width of $(T, \mathcal{V})$, i.e. at most $\text{tw}(G)$. $\square$

A tree-decomposition $(T, \mathcal{V})$ of G with $\mathcal{V} = (V_t)_{t \in T}$ is *linked*, or *linked/lean*, if it satisfies the following condition:

- (T4) Given $t_1, t_2 \in T$ and vertex sets $Z_1 \subseteq V_{t_1}$ and $Z_2 \subseteq V_{t_2}$ such that $|Z_1| = |Z_2| =: k$, either G contains k disjoint Z_1 - Z_2 paths or there exists an edge $tt' \in t_1 T t_2$ with $|V_t \cap V_{t'}| < k$.

The ‘branches’ in a lean tree-decomposition are thus stripped of any bulk not necessary to maintain their connecting qualities. Indeed, if a branch is thick (i.e. the adhesion sets $V_t \cap V_{t'}$ along a path in T are all large), then G is highly connected along this branch, and the bags themselves are no larger than their ‘external connectivity’ in G requires: for $t_1 = t_2$, (T4) says that two k -sets of vertices will only lie in a common bag if they cannot be separated (in G) by fewer than k vertices.

In our quest for tree-decompositions into ‘small’ bags, we now have two criteria to choose between: the global ‘worst case’ criterion of width, and the more subtle local criterion of leanness. Surprisingly, it is always possible to find a tree-decomposition that is optimal with respect to both criteria at once:

Theorem 12.4.5. (Thomas 1990)

Every graph G has a lean tree-decomposition of width $\text{tw}(G)$.

Another natural feature one might ask of a tree-decomposition is that its parts, as induced subgraphs of G , be connected. Let us call such a tree-decomposition *connected*, and let the *connected tree-width* $\text{ctw}(G)$ of G be the least width of a connected tree-decomposition of G .

The connected tree-width of most graphs is greater than their ordinary tree-width. For example, every cycle has tree-width 2, but $\text{ctw}(C^n) = \lceil n/2 \rceil$ (Exercise 33). And unlike ordinary tree-width, the connected tree-width of a subgraph of G can be greater than that of G . (For example, let G be obtained from a long cycle by adding a chord between opposite vertices.) However, if $C \subseteq G$ is a *geodesic* cycle, i.e., if $d_C(u, v) = d_G(u, v)$ for all vertices $u, v \in C$, then $\text{ctw}(C) \leq \text{ctw}(G)$.

The presence of long geodesic cycles in a graph thus is an obstruction to small connected tree-width – as is, trivially, large ordinary tree-width. By the following theorem, however, these are the only obstructions:

Theorem 12.4.6. *There is a function $f: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that every graph of tree-width $\leq w \in \mathbb{N}$ that has no geodesic cycle of length $> k \in \mathbb{N}$ has connected tree-width at most $f(w, k)$.*

12.5 Tangles

We have already in this chapter met a few types of substructures of possibly sparse graphs that are highly connected in some sense, but which are not just k -connected subgraphs (or minors) for some large k . Large grid minors are an example, the k -blocks defined at the end of Section 12.3 are another, as are brambles of high order. All these substructures have one feature in common: for every low-order separation of the graph they lie essentially, though not necessarily entirely, on one of its two sides.

For example, given a bramble $\mathcal{B}$ of order k in a graph $G = (V, E)$, if $\{A, B\}$ is a separation of order $< k$ then $A \setminus B$ or $B \setminus A$, but not both, contains a set from $\mathcal{B}$. This helped us prove the easy implication of the tree-width duality theorem: $\mathcal{B}$ orients every edge of the decomposition tree T of any tree-decomposition $(T, \mathcal{V})$ of adhesion $< k$ ‘towards’ the